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Student Number

2016

Mathematics Extension 2

HSC Course Assessment Task 2

Date: 15/03/16

General Instructions

- Reading time – 5 minutes
- Working time 3 hours
- Write using blue or black pen
Black pen is preferred
- Approved calculators may be used
- In Questions 11 – 16 show relevant mathematical reasoning and/or calculations
- Answer each question in a separate writing booklet
- This paper must not be removed from the examination room
- A reference sheet is provided at the back of this paper
- Diagrams are NOT to scale

Total Marks – 100

Section I - Pages 2 - 4

10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II - Pages 5 - 10

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes for this section

QUESTION 1-10 Multiple Choice	/10
QUESTION 11	/15
QUESTION 12	/15
QUESTION 13	/15
QUESTION 14	/15
QUESTION 15	/15
QUESTION 16	/15

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 30% of the Preliminary / HSC Course Assessment

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for questions 1 – 10 (Detach from paper)

- 1 Consider the curve: $\frac{x^2}{a} + \frac{y^2}{b} = 1$, which of the following statement is true?
- (A) Every line passing through the origin is a line of symmetry
 - (B) The x -intercepts are $(a, 0)$ and $(-a, 0)$
 - (C) The y -intercepts are $(0, \sqrt{b})$ and $(0, -\sqrt{b})$
 - (D) The curve is an ellipse centre at (a, b)
- 2 Find all solutions of $z^3 = -1$
- (A) $z = -1$
 - (B) $z = -1, \frac{1 \pm \sqrt{3}i}{2}$
 - (C) $z = -1, \frac{1 \pm \sqrt{3}}{2}$
 - (D) $z = \pm 1,$
- 3 When a plane intersects a cone at an angle that is parallel to the slant edge of the cone, what shape is formed?
- (A) Parabola
 - (B) Hyperbola
 - (C) Circle
 - (D) Ellipse

- 4 Evaluate $\frac{i}{5-i}$
- (A) $\frac{-1+5i}{26}$
(B) $\frac{5-i}{26}$
(C) $-\frac{1}{5}+i$
(D) None of the above
- 5 Which of the following is an equation of an ellipse with center $(2, -3)$, horizontal major axis of length 10 and minor axis of length 6.
- (A) $\frac{(x-2)^2}{25} + \frac{(y+3)^2}{9} = 1$
(B) $\frac{(x+2)^2}{5} + \frac{(y-3)^2}{3} = 1$
(C) $\frac{(x-2)^2}{10} + \frac{(y+3)^2}{6} = 1$
(D) $\frac{(x-2)^2}{10^2} + \frac{(y+3)^2}{6^2} = 1$
- 6 Consider the following equation: $|x + 1| + |x - 1| = 2$
- (A) It has only 1 solution
(B) It has 2 solutions
(C) It has no solutions
(D) None of the above
- 7 The equation $x^3 + 2x - 1 = 0$ has roots α, β and γ . Find the monic equation with roots α^2, β^2 and γ^2 .
- (A) $x^3 + 2x^2 + 2x - 3 = 0$
(B) $x^3 + 4x^2 + 2x - 1 = 0$
(C) $x^3 + 4x^2 + 4x - 1 = 0$
(D) $x^3 + 4x^2 + 4x - 3 = 0$

- 8 Let $z = \sqrt{48} - 4i$. What is the value $\arg(z^7)$?
- (A) $\frac{-2\pi}{3}$
- (B) $\frac{2\pi}{3}$
- (C) $\frac{5\pi}{6}$
- (D) $\frac{-5\pi}{6}$
- 9 Given the graph of $y = f(x)$, how would you sketch the graph of $y = f(2 - x)$.
- (A) Shift $y = f(x)$ to the right by 2 units and then reflect on y -axis
- (B) Shift $y = f(x)$ to the left by 2 units and then reflect on x -axis
- (C) Reflect on y -axis and then shift to the left by 2 units
- (D) Reflect on y -axis and then shift to the right by 2 units
- 10 The polynomial $P(x) = x^5 + ax^4 + x^3 + bx^2 - 8x - 4$ has a double root at $x = -1$. What are the values of a and b ?
- (A) $a = -2$ and $b = -4$
- (B) $a = 2$ and $b = -4$
- (C) $a = 2$ and $b = 4$
- (D) $a = -2$ and $b = 4$

End of Section I

Section II

90 marks

Attempt Questions 11 – 15

Allow about 2 hours and 15 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 15, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet

- (a) Let $z = 2i + 1$ and $w = 5 - 3i$, evaluate the following and give your answer in the form of $a + ib$. **1**
- (i) $z + 2w$ **2**
- (ii) $\frac{z}{w}$ **2**
- (iii) $z\bar{w} + \bar{z}w$
- (b) Show that $\frac{1+2i}{3-4i} + \frac{2-i}{5i}$ is a real number. **2**
- (c) Let ω be a non real cube root of unity.
- (i) Show that $\omega^2 + \omega + 1 = 0$ **2**
- (ii) Hence show that: $(b + c)(b + c\omega)(b + c\omega^2) = b^3 + c^3$ **2**
- (d) (i) Express $z = \frac{i-1}{\sqrt{3}+i}$ in mod-arg form. **2**
- (ii) Hence evaluate $\cos\left(\frac{7\pi}{12}\right)$ in surd form. **2**

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet

- (a) A cubic equation: $x^3 - 2x^2 + 13x - 12 = 0$ has roots α, β and γ . Given that:

$$\alpha^2 + \beta^2 + \gamma^2 = -22$$

Evaluate $\alpha^3 + \beta^3 + \gamma^3$

2

- (b) The roots of $x^3 + x + 1 = 0$ are α, β, γ . Find the cubic equation whose roots are: $\frac{1}{1-\alpha}, \frac{1}{1-\beta}, \frac{1}{1-\gamma}$

2

- (c) (i) Use De Moivre's theorem to express $\sin(3\theta)$ in term of $\sin(\theta)$. **2**
- (ii) Use the results above to solve: $8x^2 - 6x + \sqrt{3} = 0$, give answer to 3 significant figures. **2**
- (iii) Hence deduce that: $\sin\left(\frac{\pi}{9}\right) + \sin\left(\frac{7\pi}{9}\right) = \sin\left(\frac{5\pi}{9}\right)$ **1**

- (d) The polynomial $P(x) = 3x^4 + 2x^3 - 6x^2 - 6x + a$ has a root of multiplicity 2.
- (i) Find the possible values of a . **3**
- (ii) If a is positive, state the repeated zero, and hence factorise $P(x)$ into linear factors. **3**

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet

- (a) $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ are points on $xy = c^2$ which moves such that P, Q and $S(c\sqrt{2}, c\sqrt{2})$ are always collinear.
- (i) Show that the tangent to $xy = c^2$ at any point $T\left(ct, \frac{c}{t}\right)$ is given by: 2
$$x + t^2y = 2ct$$
- (ii) Show that the tangents at P and Q intersect at point $R: \left(\frac{2cpq}{p+q}, \frac{2c}{p+q}\right)$ 1
- (iii) Show that $p + q = \sqrt{2}(1 + pq)$ 1
- (iv) Find the locus of R 1
- (b) The tangent at $P(x, y)$ to the hyperbola $x^2 - y^2 = a^2$ in the first quadrant meets the x -axis at acute angle α . The line OP , where O is the origin, meets the x -axis at an acute angle β .
- (i) Prove that the product of the gradients of the line OP and tangent at P is 1. 1
- (ii) Show that $\alpha + \beta = \frac{\pi}{2}$ 2
- (c) (i) Show that the equation of the chord joining the points $P(cp, c/p)$ and $Q(cq, c/q)$ is $x + pqy = c(p + q)$. 2
- (ii) Hence determine the gradient of the tangent to the rectangular hyperbola $xy = c^2$ at the point $P(cp, c/p)$. 1
- (iii) The chord AB of the hyperbola $xy = c^2$ subtends a right angle at a point P on the curve.
- (α) Draw a diagram to show the information 1
- (β) Prove that AB is parallel to the normal at P . 3

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

- (a) $\bar{z}$ is the complex conjugate of z . If $z\bar{z} + 2iz = 12 + 6i$, find the possible solutions for z . 2

- (b) (i) If $z = r(\cos\theta + i\sin\theta)$, show, without applying De Moivre's theorem, that $z^2 = r^2(\cos 2\theta + i\sin 2\theta)$ 1

- (ii) Hence, find the square roots of $(2\sqrt{3} - 2i)$. 2

- (c) In an Argand diagram, the points P and Q represent the complex numbers $a + ib$ and $-i(a + ib)$ respectively, where a and b are real and $b > a > 0$.

- (i) On a single diagram, sketch the loci of z defined by the following equations: 3

$$|z - a - ib| = |z - b + ia|, \arg(z - a - ib) = \arg(b - ia)$$

Indicate clearly on your diagram, the point of intersection R of the two loci.

- (ii) Hence find, in terms of a and b , the complex number represented by R . 2

- (d) (i) If $z = \cos\theta + i\sin\theta$, show that 2

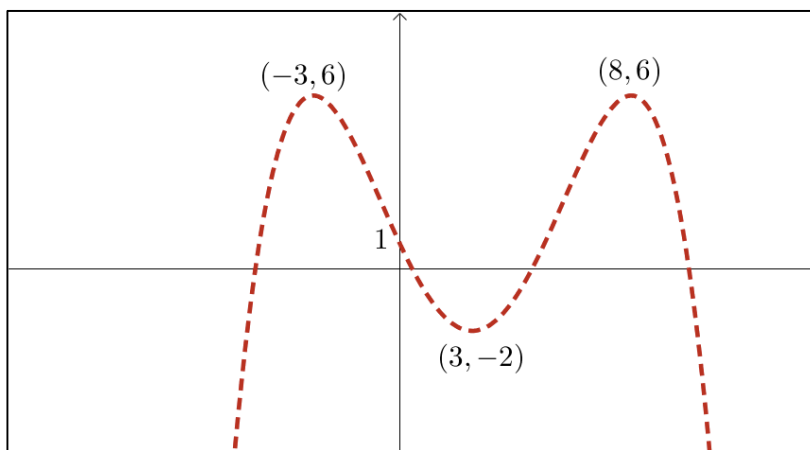
$$z^n + \frac{1}{z^n} = 2\cos(n\theta) \text{ and } z^n - \frac{1}{z^n} = 2i\sin(n\theta)$$

- (ii) Use the result to deduce that $\cos^4\theta + \sin^4\theta = \frac{1}{4}(\cos 4\theta + 3)$. 3

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet

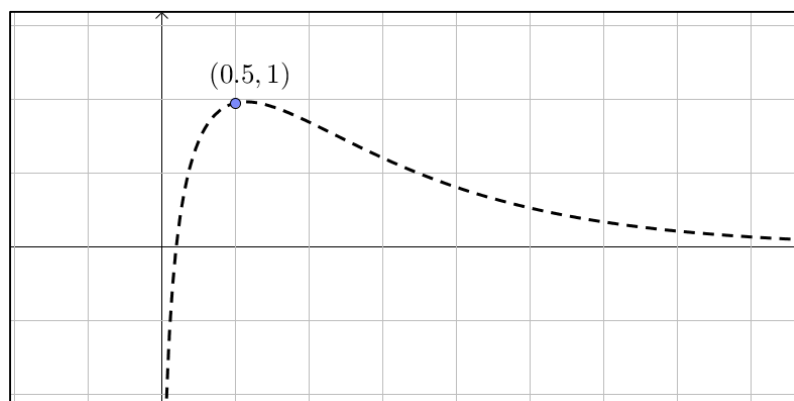
(a)



Given $y = f(x)$ in the diagram above, sketch the following graphs with $y = f(x)$ on the same axis.

- | | |
|-----------------------|----------|
| (a) $y = f'(x)$ | 1 |
| (b) $ y = f(x)$ | 2 |
| (c) $y = \sqrt{f(x)}$ | 2 |
| (d) $y = f(x)$ | 2 |

(b)



Given $y = f(x)$ in the diagram above, sketch the following graphs with $y = f(x)$ on the same axis.

- | | |
|--------------------------|----------|
| (a) $y = 3 - f(x + 1)$ | 2 |
| (b) $y = \frac{1}{f(x)}$ | 2 |
| (c) $y = 1 - f(x) $ | 2 |
| (d) $y = f(x)^2$ | 2 |

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet

- (a) (i) Show that if $y = mx + k$ is a tangent to the hyperbola: **2**
 $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then $m^2 a^2 - b^2 = k^2$.
- (ii) Hence find the equations of the tangents from the point $(1, 3)$ to **2**
the hyperbola, $\frac{x^2}{4} - \frac{y^2}{15} = 1$ and the co-ordinates of the points of
contact.
- (b) (i) Show that the parametric coordinate of the point **2**
 $P \left(\frac{a}{2} \left(t + \frac{1}{t} \right), \frac{b}{2} \left(t - \frac{1}{t} \right) \right)$ lies on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
- (ii) Show that the equation of the tangent at P in part (i) is given by **3**
$$\left(\frac{t^2 + 1}{a} \right) x - \left(\frac{t^2 - 1}{b} \right) y = 2t$$
- (iii) If S is a focus and if the tangent at P , the line through S **3**
perpendicular to the x -axis and one of the asymptotes are
concurrent. Prove that the line joining P to the focus is parallel
to the other asymptote.

End of Question 16

End of paper

2016 Killara Ext 2 Task 3 Solutions

Multiple Choice:

Q.1) C

Q.6) D

Q.2) B

Q.7) C

Q.3) A

Q.8) C

Q.4) A

Q.9) D

Q.5) A

Q.10) B

Question 11:

(a) $z = 2i + 1, w = 5 - 3i$

(i) $z + 2w$

$$= 2i + 1 + 2(5 - 3i)$$

$$= 2i + 1 + 10 - 6i$$

$$= 11 - 4i$$

1 mark: correct answer

(ii) $\frac{z}{w} = \frac{2i+1}{5-3i} \times \frac{5+3i}{5+3i}$

1 mark: multiply conjugate

$$= \frac{10i - 6 + 5 + 3i}{25 + 9}$$

$$= \frac{-1 + 13i}{34}$$

$$= -\frac{1}{34} + \frac{13}{34}i$$

1 mark: correct answer

(iii) $z\bar{w} + \bar{z}w$

1 mark: correct substitution

$$= (2i + 1)(5 + 3i) + (-2i + 1)(5 - 3i)$$

$$= 10i - 6 + 5 + 3i - 10i - 6 + 5 - 3i$$

$$= -2$$

1 mark: correct answer

(b) $\frac{1+2i}{3-4i} + \frac{2-i}{5i}$

$$= \frac{5i + 10i^2 + 6 - 3i - 8i + 4i^2}{15i + 20}$$

1 mark: common denominator

$$= \frac{-6i - 8}{15i + 20}$$

$$= \frac{-2(3i+4)}{5(3i+4)}$$

1 mark: correct answer

$$= -\frac{2}{5}$$

(c) $w^3 = 1$

1 mark: using root of unity

1 mark: reasonable explanation

(i) $w^3 - 1 = 0$

$$(w - 1)(w^2 + w + 1) = 0$$

since w is a non-real root, hence $w \neq 1$

$$\text{Therefore } (w^2 + w + 1) = 0$$

(ii) $LHS = (b + c)(b + cw)(b + cw^2)$

$$= (b + c)(b^2 + bcw^2 + bcw + c^2w^3)$$

$$= (b + c)(b^2 + bc(w^2 + w) + c^2w^3)$$

$$= (b + c)(b^2 - bc + c^2)$$

$$= b^3 - b^2c + bc^2 + b^2c - bc^2 + c^3$$

$$= b^3 + c^3 = RHS$$

1 mark: attempt to expand
1 mark: correct proof

(d) $z = \frac{i-1}{\sqrt{3}+i}$

(i) $i - 1 = \sqrt{2}Cis\left(\frac{3\pi}{4}\right)$

1 mark:

convert to mod-arg form

$$\sqrt{3} + i = 2Cis\left(\frac{\pi}{6}\right)$$

1 mark: correct answer

$$\frac{i-1}{\sqrt{3}+i} = \frac{\sqrt{2}}{2}Cis\left(\frac{3\pi}{4} - \frac{\pi}{6}\right) = \frac{\sqrt{2}}{2}Cis\left(\frac{7\pi}{12}\right)$$

$$= \frac{\sqrt{2}}{2} \left[\cos\left(\frac{7\pi}{12}\right) + i\sin\left(\frac{7\pi}{12}\right) \right]$$

(ii) $\frac{i-1}{\sqrt{3}+i} \times \frac{\sqrt{3}-i}{\sqrt{3}-i} = \frac{\sqrt{3}i - \sqrt{3} + 1 + i}{3 + 1} = \frac{1 - \sqrt{3}}{4} + \frac{1 + \sqrt{3}}{4}i$

Equating real parts:

1 mark:

attempt to equate real parts

$$\frac{\sqrt{2}}{2} \cos\left(\frac{7\pi}{12}\right) = \frac{1 - \sqrt{3}}{4}$$

1 mark: correct answer

$$\cos\left(\frac{7\pi}{12}\right) = \frac{1 - \sqrt{3}}{2\sqrt{2}} = \frac{\sqrt{2}(1 - \sqrt{3})}{4}$$

Q12

$$x^3 - 2x^2 + 13x - 12 = 0$$

wh. α, β, γ

$$\alpha^3 = 2\alpha^2 - 13\alpha + 12$$

$$\beta^3 = 2\beta^2 - 13\beta + 12$$

$$\gamma^3 = 2\gamma^2 - 13\gamma + 12$$

$$\begin{aligned} \alpha^3 + \beta^3 + \gamma^3 &= 2(\alpha^2 + \beta^2 + \gamma^2) - 13(\alpha + \beta + \gamma) + 36 \\ &= 2 \times 22 - 13 \times 2 + 36 \\ &= -34 \end{aligned}$$

b) let $x = \frac{1}{1-\alpha} \Rightarrow 1-\alpha = \frac{1}{x} \Rightarrow \alpha = 1 - \frac{1}{x} = \frac{x-1}{x}$ sub into $P(x)$

$$\left(\frac{x-1}{x}\right)^3 + \frac{x-1}{x} + 1 = 0$$

$$x^3 - 3x^2 + 3x - 1 + x^3 - x^2 + x^3 = 0$$

$$3x^3 - 4x^2 + 3x - 1 = 0 \text{ is the required equation.}$$

(c) let $z = \cos \theta + i \sin \theta$

$$z^3 = \cos 3\theta + i \sin 3\theta$$

$$= (\cos \theta + i \sin \theta)^3$$

$$= \cos^3 \theta + 3\cos^2 \theta i \sin \theta + 3\cos \theta i^2 \sin^2 \theta - i \sin^3 \theta$$

$$= \cos^3 \theta - 3\cos \theta \sin^2 \theta + [3(1 - \sin^2 \theta) \sin \theta - \sin^3 \theta] i$$

equate into parts of ① & ②

$$\Rightarrow \sin 3\theta = 3\sin \theta - 4\sin^3 \theta$$

Now $8x^3 - 6x = -\sqrt{3}$ let $x = \sin \theta$

$$\Rightarrow 8\sin^3 \theta - 6\sin \theta = -\sqrt{3} \quad \div -2$$

$$-4\sin^3 \theta + 3\sin \theta = \frac{\sqrt{3}}{2}$$

$$\text{ie } \sin 3\theta = \frac{\sqrt{3}}{2} \Rightarrow$$

$$3\theta = \frac{\pi}{3} + 2n\pi$$

$$\theta = \frac{\pi}{9} + \frac{2n\pi}{3} \Rightarrow \theta = \frac{\pi}{9}, \frac{7\pi}{9}, \frac{13\pi}{9}$$

$$\sum \text{roots} = \frac{\pi}{9} + \frac{7\pi}{9} + \frac{13\pi}{9} = 0$$

$$\text{but } \sin \frac{13\pi}{9} = \sin \left(-\frac{5\pi}{9}\right)$$

$$\therefore \frac{\pi}{9} + \frac{7\pi}{9} = \frac{8\pi}{9}$$

$$\begin{aligned} \theta &= 0.342, \\ &0.643, \\ &-0.985 \end{aligned}$$

Q12(d) (i)

$$P(x) = 3x^4 + 2x^3 - 6x^2 - 6x + a$$

$P(x)$ has a root of multiplicity 2 (at say $x=1$)

$$\Rightarrow P(1) = 0 \text{ and } P'(1) = 0$$

$$\Rightarrow \left. \begin{aligned} 3x^4 + 2x^3 - 6x^2 - 6x + a &= 0 & \text{--- (1)} \\ 12x^3 + 6x^2 - 12x - 6 &= 0 & \text{--- (2)} \end{aligned} \right\} \text{1 mark}$$

$$\textcircled{2} \Rightarrow 2x^3 + x^2 - 2x - 1 = 0$$

$$x^2(2x+1) - (2x+1) = 0$$

$$\left. \begin{aligned} x=1, x=-1, x=-\frac{1}{2} \\ a=7, a=-1, a=-\frac{25}{16} \end{aligned} \right\}$$

1 mark for solving for x

1 mark for all correct values of a

$$(ii) a > 0 \Rightarrow a = 7$$

when $a=7$, the repeated root is 1 (point (i))

$\Rightarrow (x-1)^2$ is a factor of $P(x)$ $\rightarrow$ 1 mark

$$\Rightarrow P(x) = 3x^4 + 2x^3 - 6x^2 - 6x + 7 = (x-1)^2(3x^2 + 8x + 7) \rightarrow \text{1 mark}$$

$$= 3(x-1)^2 \left[x - \left(\frac{4}{3} - \frac{\sqrt{5}}{3}i \right) \right] \left[x - \left(\frac{4}{3} + \frac{\sqrt{5}}{3}i \right) \right] \rightarrow \text{1 mark}$$

2016 Killara Ext 2 Task 3 Solutions

Question 13:

(a) Given $xy = c^2$,

(i) $x \frac{dy}{dx} + y = 0 \Rightarrow \frac{dy}{dx} = -\frac{y}{x}$ **1 mark:**
correctly differentiate

At $T \left(ct, \frac{c}{t} \right)$, gradient is $-\frac{1}{t^2}$

$$y - \frac{c}{t} = -\frac{1}{t^2}(x - ct)$$

$$t^2y - ct = -x + ct$$

$$x + t^2y = 2ct$$

1 mark:
correct answer

(ii) Tangent at P: $x + p^2y = 2cp$

Tangent at Q: $x + q^2y = 2cq$

The intersection R:

$$2cp - p^2y = 2cq - q^2y$$

$$2c(p - q) = (p^2 - q^2)y$$

$$y = \frac{2c}{(p+q)}$$
 1 mark:
reasonable proof

$$x = 2cp - \frac{2c}{p+q} = \frac{2cpq}{p+q}$$

(iii) Since P, Q and S are collinear:

$$M_{PQ} = M_{QS}$$

$$\frac{\frac{c}{p} - \frac{c}{q}}{cp - cq} = \frac{\frac{c}{q} - c\sqrt{2}}{cq - c\sqrt{2}}$$

$$-\frac{q - \sqrt{2}}{pq} = \frac{1}{q} - \sqrt{2}$$

$$-q + \sqrt{2} = p - pq\sqrt{2}$$

$$\sqrt{2} + pq\sqrt{2} = p + q$$

$$\sqrt{2}(1 + pq) = p + q$$
 1 mark:
reasonable proof

(iv) $x = \frac{2cpq}{p+q}$ $y = \frac{2c}{p+q}$

$$x = \frac{2c(pq)y}{2c}$$

$$x = y \left(\frac{p+q}{\sqrt{2}} - 1 \right)$$

$$x = y \left(\frac{2c}{y\sqrt{2}} - 1 \right)$$

$$x = \frac{2c}{\sqrt{2}} - y$$

$$x = \sqrt{2}c - y$$

$$y = \sqrt{2}c - x$$

1 mark:
correctly eliminate
the parameters

(b) Given $x^2 - y^2 = a^2$

(i) $2x - 2y \frac{dy}{dx} = 0$

$$\frac{dy}{dx} = \frac{x}{y}$$

Slope at P: $\frac{x}{y}$

1 mark:
reasonable proof

Slope OP: $\frac{y-0}{x-0} = \frac{y}{x}$

$$M_P \times M_{OP} = \frac{x}{y} \times \frac{y}{x} = 1$$

(ii) Since $\tan(\alpha) = \frac{x}{y}$ and $\tan(\beta) = \frac{y}{x}$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$= \frac{\tan \alpha + \tan \beta}{1 - 1} \quad (\text{undefined})$$

Therefore $\alpha + \beta = \frac{\pi}{2}$

1 mark:
decent attempt

1 mark:
reasonable explanation

Q13(c)(i) $P(cp, cq), Q(cq, cq)$

gradient of $PQ = \frac{cq - cq}{cq - cp}$

$= \frac{p - q}{pq(q - p)}$

$= -\frac{1}{pq}$

→ 1 mark for getting the gradient

∴ Eqⁿ of PQ is

$y - \frac{c}{p} = -\frac{1}{pq}(x - cp)$

⇒ $x + pqy = c(p+q)$

→ 1 mark

(ii) as $Q \rightarrow P$, chord $PQ \rightarrow$ tangent at P

⇒ $q = p$

⇒ Eqⁿ of tangent at P is,

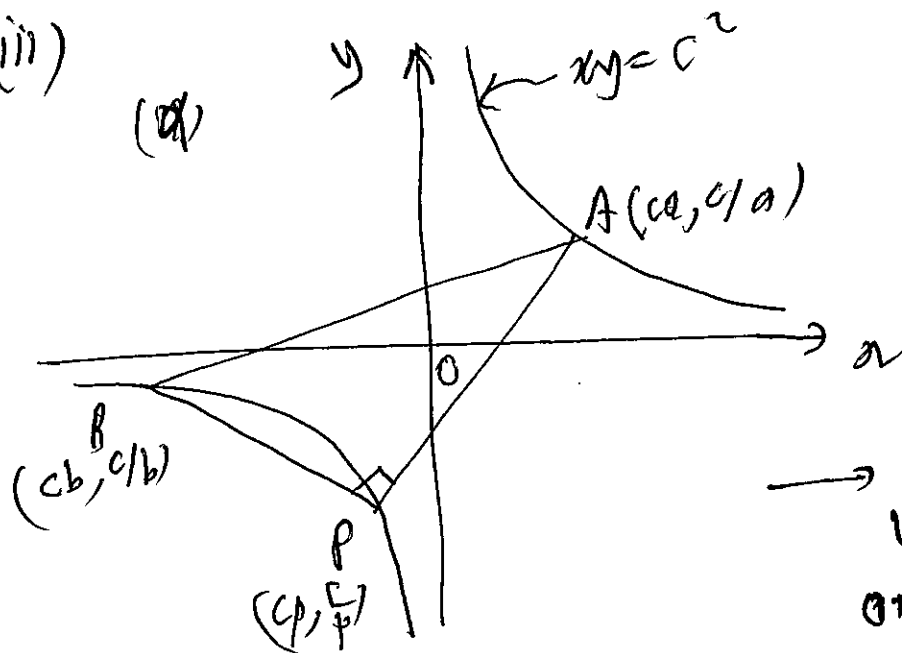
$x + p^2y = c(p+p)$

(ie) $x + p^2y = 2cp$

→ 1 mark

~~Q14~~

Q, 3 (iii)



→ 1 mark for
Line A and B
on diff branch
of the hyperbola

(b) $\angle APB = 90^\circ$

$\therefore m_{AP} \times m_{BP} = -1$

→ 1 mark

$\Rightarrow -\frac{1}{ap} \times -\frac{1}{bp} = -1$

$\Rightarrow p^2 = -\frac{1}{ab}$

→ 1 mark

gradient of the normal at P is p^2 from c(ii), above
gradient of the chord AB is $-\frac{1}{ab}$ from c(i), above

→ 1 mark

$\Rightarrow$ normal at $P \parallel$ chord AB

Q14. (a) $z\bar{z} + 2iz = 12 + 6i$

let $z = a + ib$

$\Rightarrow (a^2 + b^2 - 2b) + 2ia = 12 + 6i$

Matching the real parts

$a^2 + b^2 - 2b = 12$

and matching the imaginary parts

$2a = 6$

1 mark for the 2 equations

$\Rightarrow \begin{cases} a = 3 \\ b = 3 \end{cases} \quad \begin{cases} a = 3 \\ b = -1 \end{cases}$

$\therefore z = \underline{3 + 3i}$ or $z = \underline{3 - i}$ → 1 mark for correct answer

(b) (i) $z = r(\cos\theta + i\sin\theta)$

$z^2 = r^2(\cos\theta + i\sin\theta)^2$

$= r^2[(\cos^2\theta - \sin^2\theta) + 2i\sin\theta\cos\theta]$

$= r^2[\cos 2\theta + i\sin 2\theta]$

1 mark

(ii) $(2\sqrt{3} - 2i) = 4 \left[\frac{\sqrt{3}}{2} - \frac{1}{2}i \right]$

$= 4 \left[\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right) \right]$

$= 4 \left[\cos\left(2\left(-\frac{\pi}{12}\right)\right) + i\sin\left(2\left(-\frac{\pi}{12}\right)\right) \right]$

$\therefore \sqrt{2\sqrt{3} - 2i} = \pm 2 \left[\cos\left(\frac{\pi}{12}\right) - i\sin\left(\frac{\pi}{12}\right) \right]$

1 mark

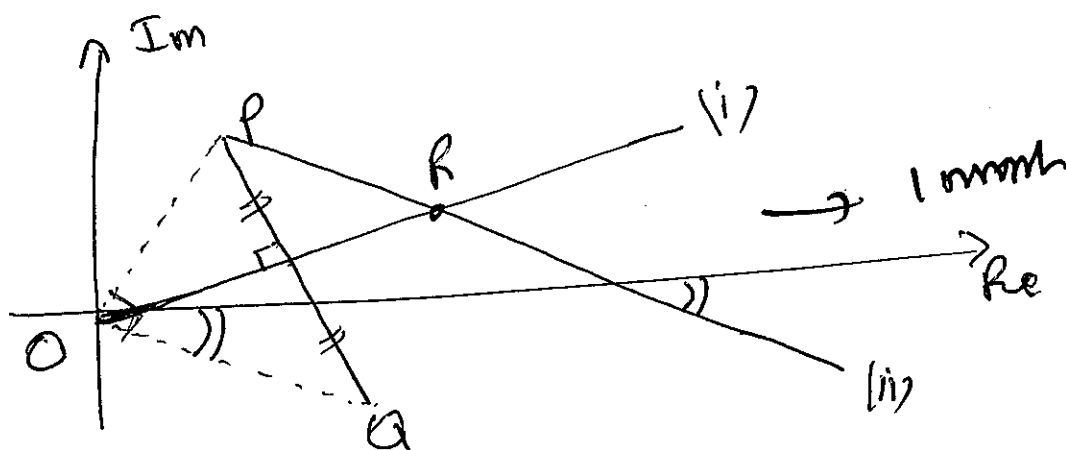
14 (c)

P represents $a+ib$, $b > a > 0$

Q represents $-i(a+ib) = b-ia$

(i) $|z-a-ib| = |z-b+ia|$
 $\equiv$ perpendicular bisector of line segment joining P and Q $\rightarrow$ 1 mark

~~arg~~ $\arg(z-a-ib) = \arg(b-ia)$
 $\equiv$ half line starting from P in the direction parallel to OQ $\rightarrow$ 1 mark



(ii) . Let $p, q, r =$ complex numbers representing by points P, Q, R

Deduce that $OPRQ$ is a square $\rightarrow$ 1 mark

$$\Rightarrow r = p + q = (a+ib) + (b-ia) \rightarrow 1 \text{ mark}$$

$$= (a+b) + i(b-a) \text{ (Answer)}$$

14) (d)

$$Z = \cos \theta + i \sin \theta \Rightarrow Z^n = \cos(n\theta) + i \sin(n\theta) \quad \text{--- (1)}$$

De Moivre's Theorem } 1 \text{ mark}

$$\text{and } \frac{1}{Z^n} = \cos(n\theta) - i \sin(n\theta) \quad \text{--- (2)}$$

$$\begin{aligned} \text{(1) + (2)} \quad Z^n + \frac{1}{Z^n} &= 2 \cos(n\theta) \\ \text{(1) - (2)} \quad Z^n - \frac{1}{Z^n} &= 2i \sin(n\theta) \end{aligned} \quad \left. \vphantom{\begin{aligned} \text{(1) + (2)} \\ \text{(1) - (2)} \end{aligned}} \right\} \rightarrow 1 \text{ mark}$$

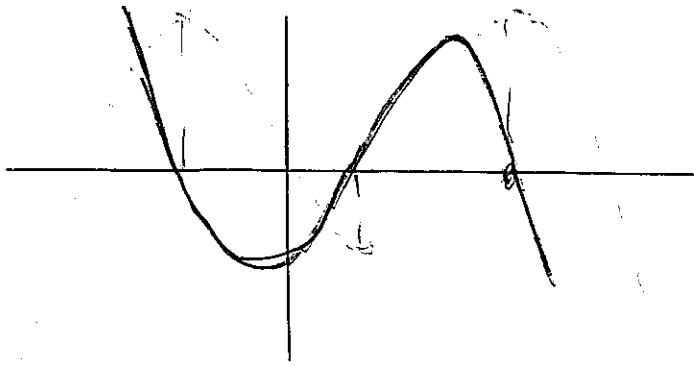
$$\begin{aligned} 2 \cos \theta &= Z + \frac{1}{Z} \\ \therefore 16 \cos^4 \theta &= \left(Z + \frac{1}{Z} \right)^4 \quad \text{--- (3)} \\ 2i \sin \theta &= Z - \frac{1}{Z} \\ \therefore 16 \sin^4 \theta &= \left(Z - \frac{1}{Z} \right)^4 \quad \text{--- (4)} \end{aligned} \quad \left. \vphantom{\begin{aligned} 16 \cos^4 \theta \\ 16 \sin^4 \theta \end{aligned}} \right\} 1 \text{ mark}$$

$$\begin{aligned} \text{(3) + (4)} \quad 16 (\cos^4 \theta + \sin^4 \theta) &= \left(Z + \frac{1}{Z} \right)^4 + \left(Z - \frac{1}{Z} \right)^4 \\ &= 2 \left(Z^4 + \frac{1}{Z^4} \right) + 12 \end{aligned} \quad \rightarrow 1 \text{ mark}$$

$$16 (\cos^4 \theta + \sin^4 \theta) = 2 \left[2 \cos 4\theta \right] + 12 \quad \left. \vphantom{16 (\cos^4 \theta + \sin^4 \theta)} \right\} 1 \text{ mark for simplifying}$$

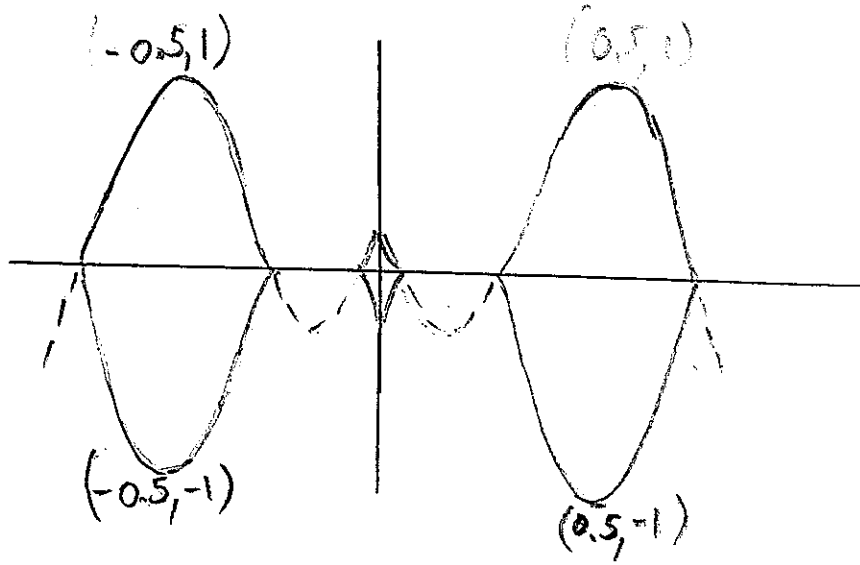
$$\therefore \cos^4 \theta + \sin^4 \theta = \frac{1}{4} [\cos 4\theta + 3] \quad \underline{\text{Ans}}$$

(a)



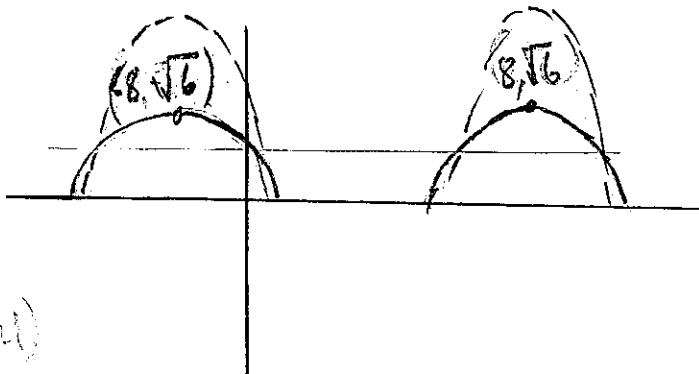
(b)

$$|y| = f(|x|)$$

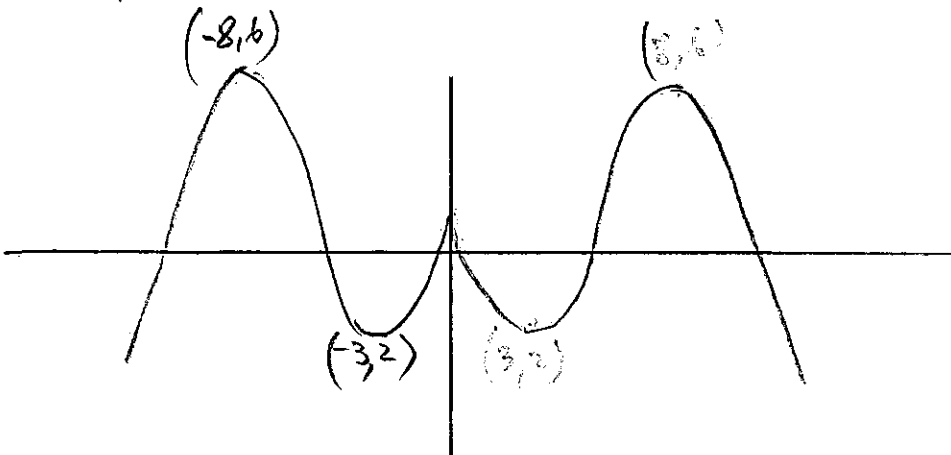


(c)

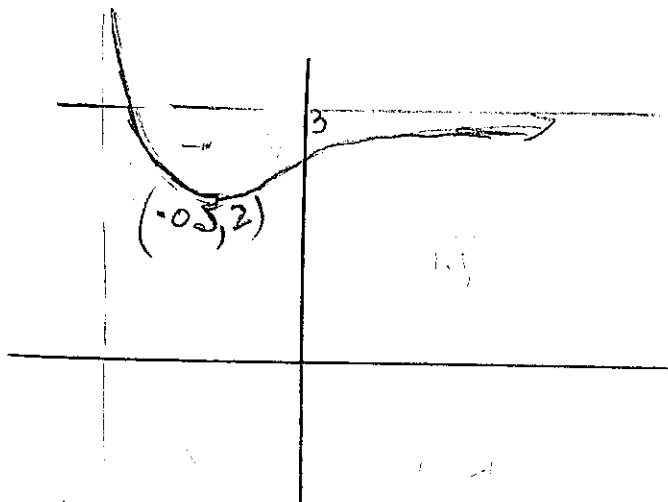
$$y = \sqrt{x}$$



(d) $y = f(|x|)$



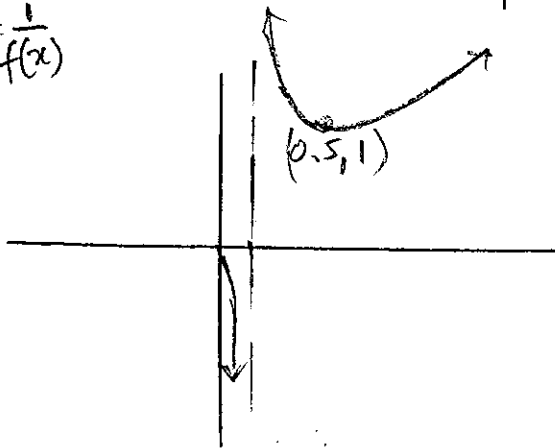
a)



$$y = 3 - f(x+1)$$

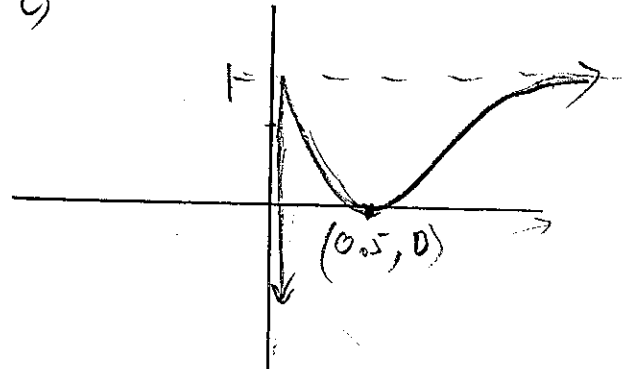
b)

$$y = \frac{1}{f(x)}$$

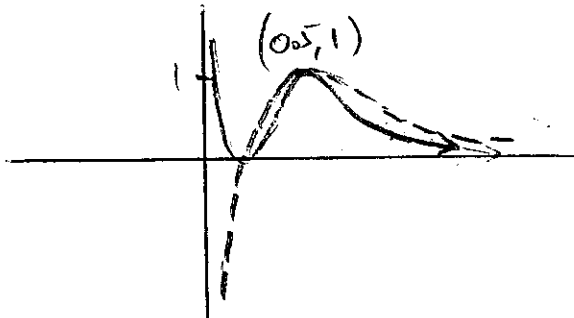


c)

$$y = 1 - |f(x)|$$



d)



$$y = [f(x)]^2$$

Q16 (a) Solve simultaneously $\begin{cases} y = mx + k \\ \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \end{cases}$ for pts. of contact.

$$\frac{x^2}{a^2} - \frac{(mx+k)^2}{b^2} = 1 \Rightarrow x^2(b^2 - a^2m^2) - 2mka^2x - a^2k^2 + a^2b^2 = 0 \quad (1)$$

Since $y = mx + k$ is the tangent to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, there is only 1 pt. of intersection $\therefore \Delta = 0$

$$\Delta = 4m^2k^2a^4 + 4(b^2 - a^2m^2)(a^2k^2 + a^2b^2) = 0$$

$$4a^2m^2b^2 = 4a^2b^2k^2 + 4a^2b^2$$

$$\Rightarrow a^2m^2 = k^2 + b^2 \Rightarrow a^2m^2 - b^2 = k^2$$

(ii) The equation of the tangents are $y = mx \pm \sqrt{a^2m^2 - b^2}$

Since $(1, 3)$ lies on the tangents, sub. $(1, 3)$, $a^2 = 4, b^2 = 15$

$$\Rightarrow (3 - m)^2 = 4m^2 - 15 \Rightarrow 9 - 6m + m^2 = 4m^2 - 15$$

$$(m + 4)(m - 2) = 0$$

$$\therefore m = -4 \text{ \& } 2$$

When $m = -4$

$$\text{Sub. into } k = \sqrt{a^2m^2 - b^2}$$

$$\Rightarrow k = \sqrt{4 \times 16 - 15} = 7$$

$$m = 2$$

$$k = \sqrt{4 \times 16 - 15} = 1$$

$\therefore$ Equations of the 2 tangents are $y = -4x + 7$
or $y = 2x + 1$

(iii) The points of contact are found by solving (1) for x .

$$\text{Since } \Delta = 0 \Rightarrow x = \frac{b}{2a}$$

$$\text{i.e. } x = \frac{2mka^2}{2(b^2 - a^2m^2)}$$

$$\text{When } \begin{cases} m = -4 \\ k = 7 \end{cases} \Rightarrow x = \frac{-2 \times 7 \times 4 \times 4}{2(15 - 4 \times 16)} = \frac{16}{7}$$

$$y = -\frac{15}{7}$$

$\therefore (\frac{16}{7}, -\frac{15}{7})$ is pt. of contact of $y = -4x + 7$

$$\text{When } \begin{cases} m = 2 \\ k = 1 \end{cases} \Rightarrow x = \frac{2 \times 2 \times 1 \times 4}{2(15 - 4 \times 4)} = -8$$

$$y = -15$$

$(-8, -15)$ is pt. of contact of $y = 2x + 1$

b) Sub $x = \frac{a}{2} \left(t + \frac{1}{t}\right)$ & $y = \frac{b}{2} \left(t - \frac{1}{t}\right)$ into $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$$\text{L.H.S} = \frac{\frac{a^2}{4} \left(t^2 + 2 + \frac{1}{t^2}\right)}{a^2} - \frac{\frac{b^2}{4} \left(t^2 - 2 + \frac{1}{t^2}\right)}{b^2} \checkmark$$

$$= \frac{4}{4} = 1 = \text{R.H.S} \quad \checkmark$$

$\therefore$ P lies on the hyperbola

(ii) gradient of the tangent :

$$\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{b^2 x}{a^2 y}$$

$$\text{at } \left(\frac{a}{2} \left(t + \frac{1}{t}\right), \frac{b}{2} \left(t - \frac{1}{t}\right)\right) \Rightarrow m = \frac{\frac{b^2 a}{2} \left(t + \frac{1}{t}\right)}{\frac{a^2 b}{2} \left(t - \frac{1}{t}\right)} = \frac{b}{a} \left(\frac{t^2 + 1}{t^2 - 1}\right) \checkmark$$

$\therefore$ equation of the tangent is :

$$y - \frac{b}{2} \left(t - \frac{1}{t}\right) = \frac{b}{a} \left(\frac{t^2 + 1}{t^2 - 1}\right) \left(x - \frac{a}{2} \left(t + \frac{1}{t}\right)\right) \checkmark$$

$$\left(\frac{t^2 + 1}{a}\right)x - \left(\frac{t^2 - 1}{b}\right)y = -\frac{b}{2} \left(1 - \frac{1}{t^2}\right) \cdot \frac{(t^2 - 1)}{b} + \frac{a}{2} \cdot \frac{t^2 + 1}{a} \left(t + \frac{1}{t}\right) \checkmark$$

$$= -\frac{(t^2 - 1)^2}{2t} + \frac{(t^2 + 1)^2}{2t}$$

$$= \frac{-t^4 + 2t^2 - 1 - t^4 + 2t^2 + 1}{2t} = 2t \checkmark$$

(iii) Sub $x = ae$, into $y = \frac{bx}{a}$

$$\Rightarrow y = \frac{bae}{a} = be \text{ ie the concurrent point } P(ae, be)$$

P lies on the tangent, Sub (ae, be) :

$$\Rightarrow \left(\frac{t^2 + 1}{a}\right)x - \left(\frac{t^2 - 1}{b}\right)y = 2t \Rightarrow (t^2 + 1)e - (t^2 - 1)e = 2t \checkmark$$

$$\Rightarrow 2e = 2t \quad \therefore \underline{e = t}$$

hence, P has coordinates $\left(\frac{a}{e} \left(e + \frac{1}{e}\right), \frac{b}{2} \left(e - \frac{1}{e}\right)\right) \checkmark$

$$m_{PS} = \frac{\frac{b}{2} \left(e - \frac{1}{e}\right)}{\frac{a}{2} \left(e + \frac{1}{e}\right) - ae} = -\frac{b(e^2 - 1)}{a(e^2 - 1)} = -\frac{b}{a} \checkmark$$

The gradient of the other asymptote is $-\frac{b}{a}$

$$\therefore m_{PS} \parallel m_y = -\frac{b}{a}$$